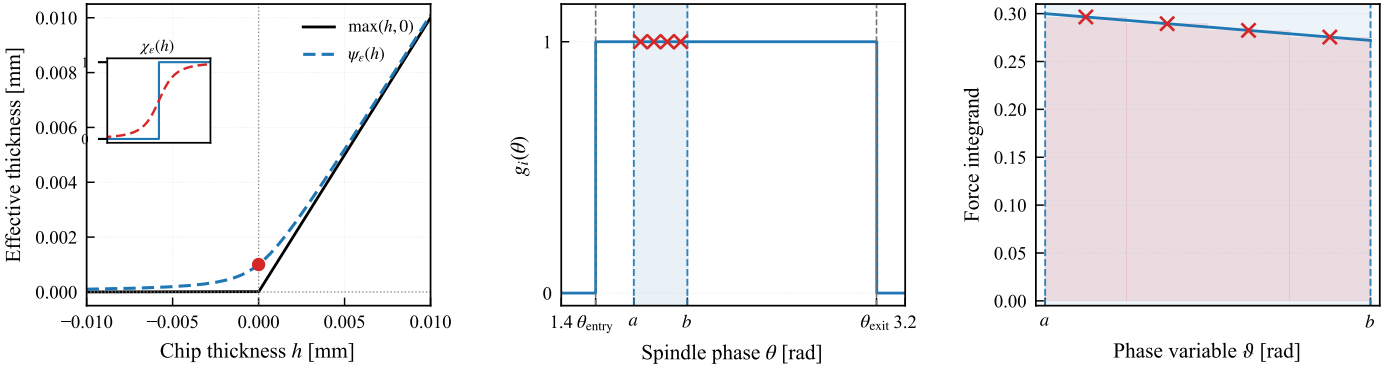
(a) Chip-loss
Effective thickness [mm]
0.010
0.008
0.006
0.004
0.002
0.000
−0.010
−0.005
0.000
0.005
0.010
Chip thickness h [mm]
$\max(h, 0)$
$\psi_\epsilon(h)$
$\chi_\epsilon(h)$
1
0
(b) Engagement
$g_i(\theta)$
1
0
1.4 θ_{entry}
a
b
θ_{exit} 3.2
Spindle phase θ [rad]
(c) Active-interval quadrature
Force integrand
0.30
0.25
0.20
0.15
0.10
0.05
0.00
a
b
Phase variable ϑ [rad]